

ROSTOV, Russian Federation

WMO: 273290

Lat: 57.200N

Lon: 39.417E

Elev: 101

StdP: 100.12

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-23.2	-28.8	0.3	-26.4	-25.5	0.4	-23.0	7.6	-2.8	6.6	-3.4	1.1	350	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.2	20.3	27.3	19.8	25.5	19.0	21.8	27.4	20.7	25.9	19.6	24.4	2.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.6	24.9	18.7	13.7	23.6	17.7	12.8	22.5	64.0	27.5	60.0	26.2	56.5	24.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.6	5.0	-29.7	30.9	3.0	1.9	-31.9	32.3	-33.7	33.5	-35.4	34.6	-37.5	36.0		
(5)			-29.8	23.0	3.0	1.4	-32.0	24.0	-33.8	24.8	-35.5	25.5	-37.6	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-8.8	-8.3	-3.0	5.3	12.2	16.2	19.0	16.3	11.1	4.8	-1.7	-6.0
	DBStd	11.05	7.27	7.32	5.11	4.94	4.85	3.83	3.46	3.62	3.83	4.41	5.44	6.94
	HDD10.0	2736	583	513	402	156	33	2	0	1	32	167	350	497
	HDD18.3	5033	842	747	660	390	198	87	33	83	219	418	600	756
	CDD10.0	852	0	0	0	16	101	187	280	197	64	8	0	0
	CDD18.3	107	0	0	0	0	7	22	55	21	2	0	0	0
	CDH23.3	961	0	0	0	2	70	204	482	194	8	0	0	0
	CDH26.7	238	0	0	0	0	15	44	124	54	0	0	0	0
Wind	WSAvg	2.2	2.4	2.5	2.5	2.3	2.3	2.0	1.8	1.8	1.9	2.3	2.5	2.6
Precipitation	PrecAvg	588	41	32	29	32	50	69	63	70	58	57	46	42
	PrecMax	815	62	74	69	64	100	159	163	184	129	139	84	77
	PrecMin	386	16	5	6	13	16	1	6	14	21	15	3	10
	PrecStd	98	13	14	14	13	17	33	35	40	28	33	22	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	3.8	11.7	22.4	28.7	30.2	32.2	31.5	24.5	18.3	10.3	7.1
		MCWB	2.4	2.0	6.2	13.3	18.6	20.2	22.2	21.9	18.4	13.4	8.6	6.1
	2%	DB	1.9	2.4	7.4	19.3	25.4	27.9	29.6	27.6	21.6	14.7	7.6	4.6
		MCWB	1.0	1.1	3.4	11.7	17.4	19.5	20.8	20.3	16.3	11.5	6.6	3.9
	5%	DB	1.2	1.6	4.9	16.3	22.7	25.8	27.8	25.2	19.4	12.5	6.2	2.7
		MCWB	0.5	0.6	2.2	10.1	15.2	18.8	20.4	19.0	15.0	10.2	5.3	1.9
	10%	DB	0.4	0.9	3.3	13.5	20.6	23.7	26.0	23.0	17.4	10.9	4.7	1.5
		MCWB	-0.2	0.2	1.4	8.5	14.1	17.6	19.6	17.6	14.0	9.3	3.9	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.5	6.9	14.0	20.6	21.8	23.5	23.0	19.0	14.1	8.9	6.3
		MCDB	3.1	3.4	11.4	21.2	27.0	27.7	29.4	30.3	23.0	17.4	10.1	7.0
	2%	WB	1.2	1.5	3.9	12.3	18.2	20.5	22.3	21.2	17.1	12.1	6.8	4.1
		MCDB	1.6	2.1	6.1	18.1	24.0	26.1	27.9	26.5	20.5	14.4	7.6	4.6
	5%	WB	0.7	0.9	2.6	10.7	16.5	19.5	21.1	19.7	15.6	10.7	5.4	2.1
		MCDB	1.1	1.5	4.4	15.6	21.4	24.6	26.5	24.1	18.6	12.2	6.0	2.5
	10%	WB	-0.2	0.2	1.7	9.1	15.0	18.2	20.1	18.3	14.3	9.3	4.0	0.9
		MCDB	0.4	0.9	3.0	13.0	18.9	22.5	24.9	21.9	16.9	10.6	4.7	1.3
Mean Daily Temperature Range	5% DB	MDBR	5.8	7.1	8.1	9.3	10.0	9.8	10.0	9.6	8.5	6.0	4.1	4.7
		MCDBR	3.7	5.2	9.6	13.3	13.1	12.7	12.5	13.2	11.8	9.2	5.5	3.7
		MCWBR	3.6	4.6	6.5	7.1	6.3	5.8	5.5	6.4	6.7	6.0	4.8	3.5
	5% WB	MCDBR	3.7	4.6	7.8	12.2	11.6	11.4	11.1	12.0	10.5	7.7	5.0	3.5
		MCWBR	3.7	4.3	5.3	6.9	6.2	5.7	5.6	6.4	6.2	5.6	4.6	3.4
Clear-Sky Solar Irradiance	taub	0.284	0.326	0.346	0.403	0.384	0.374	0.412	0.417	0.372	0.350	0.314	0.278	
	taud	2.088	2.063	2.088	2.143	2.334	2.381	2.300	2.269	2.369	2.337	2.210	2.078	
	Ebn at Noon	557	687	788	802	852	866	820	784	769	674	524	448	
	Edh at Noon	57	91	116	130	114	111	118	114	89	71	52	44	
All-Sky Solar Radiation	RadAvg	0.51	1.36	2.90	4.04	5.30	5.46	5.38	4.21	2.57	1.15	0.42	0.26	
	RadStd	0.09	0.23	0.28	0.40	0.45	0.51	0.62	0.41	0.37	0.20	0.07	0.07	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.95	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page